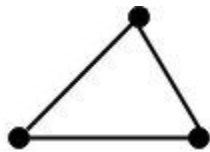
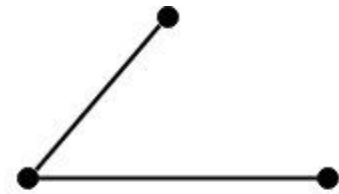


Spanning Tree

A subgraph T of a connected graph G is called spanning tree of G if T is a tree and T include all vertices of G .



Cycle Graph



Subgraph/Spanning Tree

Minimum Spanning Tree:

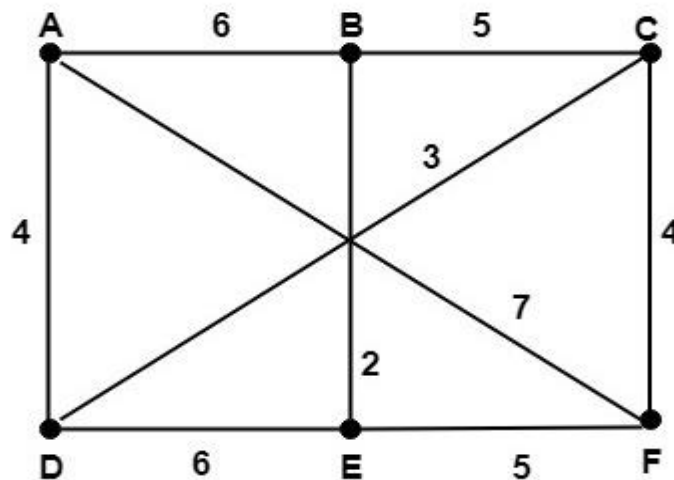
Suppose G is a connected weight graph i.e., each edge of G is assigned a non-negative number called the weight of edge, then any spanning tree T of G is assigned a total weight obtained by adding the weight of the edge in T .

A minimum spanning tree of G is a tree whose total weight is as small as possible.

Kruskal's Algorithm to find a minimum spanning tree: This algorithm finds the minimum spanning tree T of the given connected weighted graph G .

1. Input the given connected weighted graph G with n vertices whose minimum spanning tree T , we want to find.
2. Order all the edges of the graph G according to increasing weights.
3. Initialize T with all vertices but do include an edge.
4. Add each of the graphs G in T which does not form a cycle until $n-1$ edges are added.

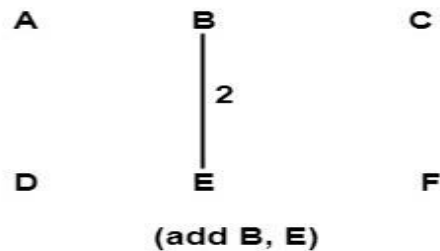
Example1: Determine the minimum spanning tree of the weighted graph shown in fig:



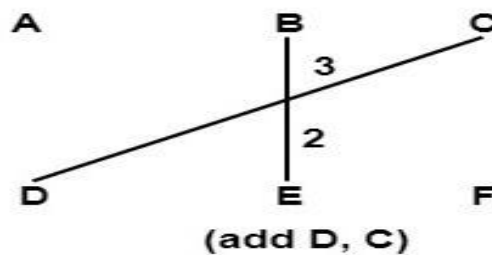
Solution: Using kruskal's algorithm arrange all the edges of the weighted graph in increasing order and initialize spanning tree T with all the six vertices of G. Now start adding the edges of G in T which do not form a cycle and having minimum weights until five edges are not added as there are six vertices.

Edges	Weights	Added or Not
(B, E)	2	Added
(C, D)	3	Added
(A, D)	4	Added
(C, F)	4	Added
(B, C)	5	Added
(E, F)	5	Not added
(A, B)	6	Not added
(D, E)	6	Not added
(A, F)	7	Not added

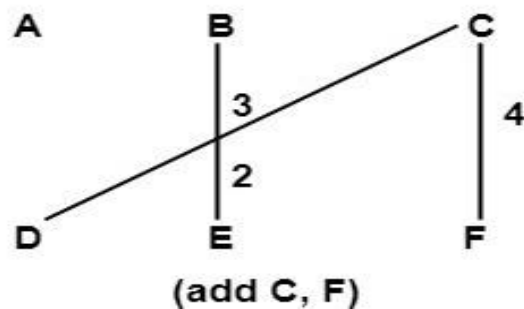
Step1:



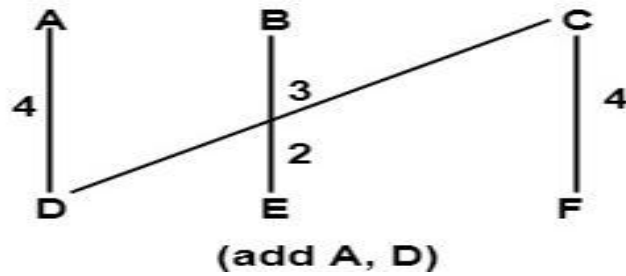
Step2:



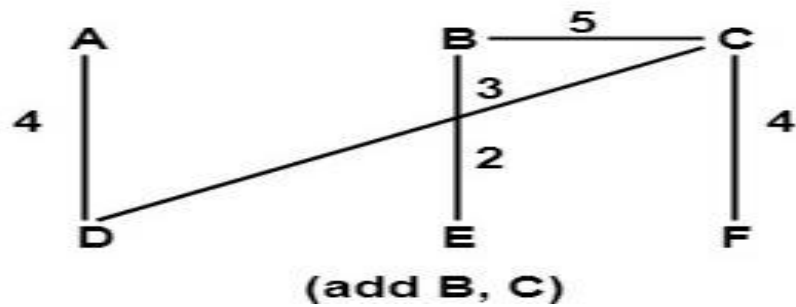
Step3:



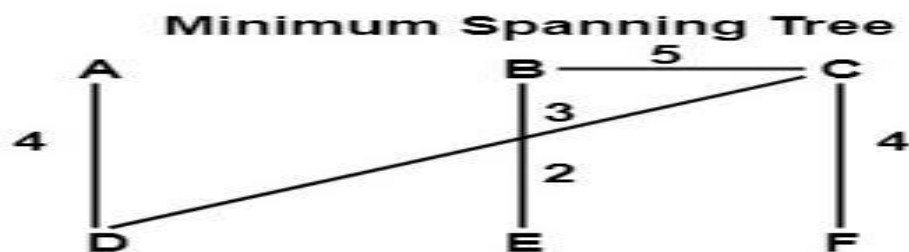
Step4:



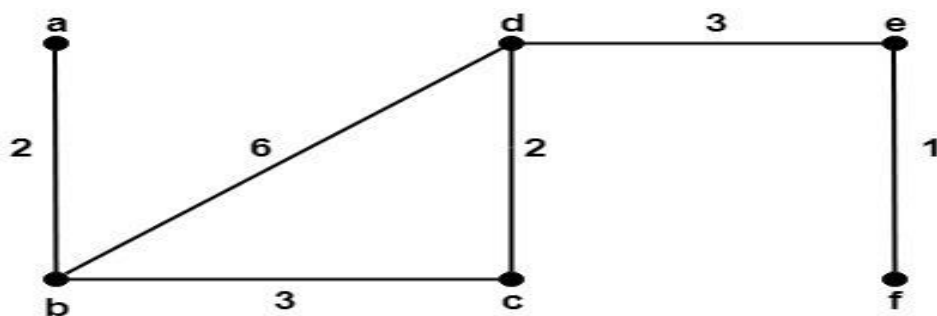
Step5:



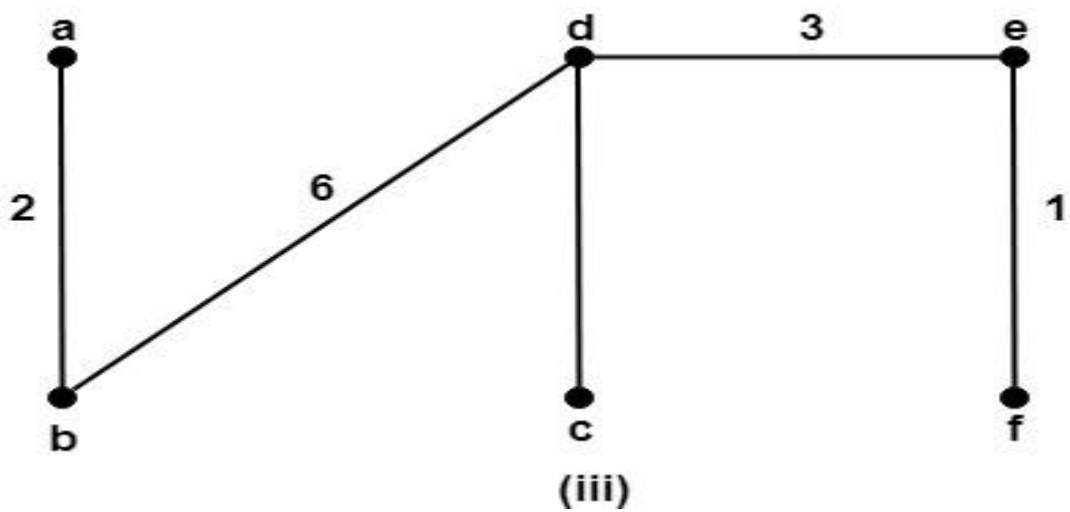
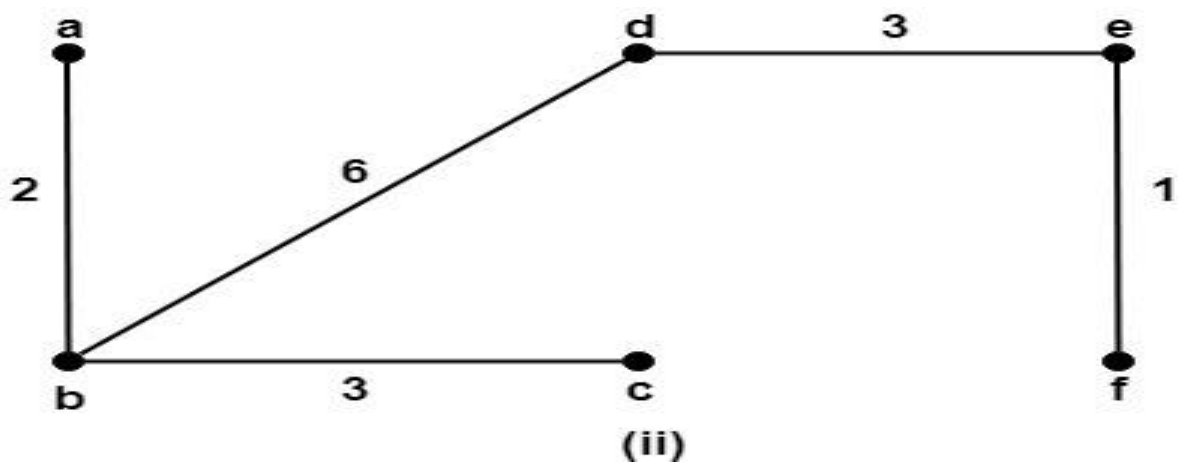
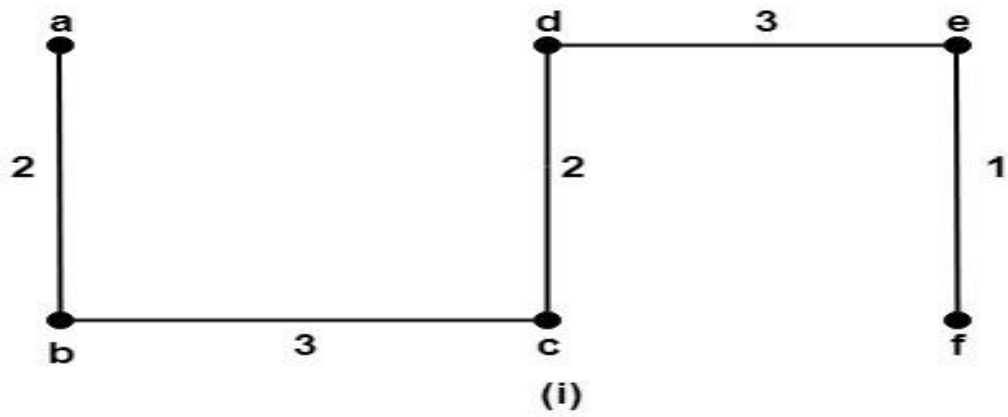
Step6: Edge (A, B), (D, E) and (E, F) are discarded because they will form the cycle in a graph. So, the minimum spanning tree form in step 5 is output, and the total cost is 18.



Example2: Find all the spanning tree of graph G and find which is the minimal spanning tree of G shown in fig:

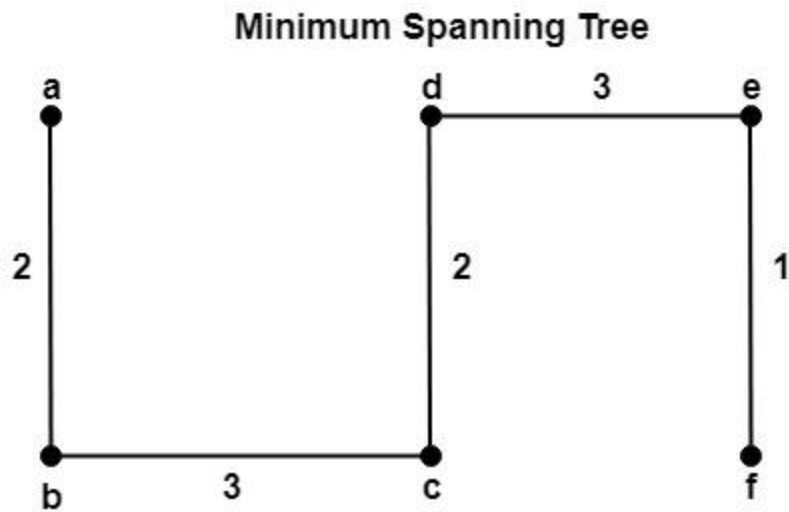


Solution: There are total three spanning trees of the graph G which are shown in fig:



To find the minimum spanning tree, use the KRUSKAL'S ALGORITHM. The minimal spanning tree is shown in fig:

Edges	Weights	Added or Not
(E, F)	1	Added
(A, B)	2	Added
(C, D)	2	Added
(B, C)	3	Added
(D, E)	3	Added
(B, D)	6	Not Added



The first one is the minimum spanning having the minimum weight = 11.